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Experiment No.: 07
Name of the Experiment: Optigrating/Design and Characterization of a Waveguide Grating.

Objective: **(A)** For a Fiber Bragg Grating operating at 1550 nm,

a) Determine the core width at which a single fiber and a single waveguide become multimode (LP Modes in fiber and TE, TM modes in waveguide). Observe mode profiles of both, fiber and waveguide.

Single Fiber Module	Single Waveguide Module
Cladding: Width = 125µm, n = 1.444 Core: Width = µm, n = 1.4475	Cladding: Width = 5µm, n = 1.444 Core: Width = µm, n = 3.473 Substrate: Width = 5µm, n = 1.444

b) What should be the grating period for a first order rectangular Bragg grating operating at a Bragg wavelength of 1.55µm for a SMF and a slab waveguide? Calculate the required period for each supported mode. **Note:** The waveguide grating should be a surface grating.

c) Observe the effect of Gaussian apodization (with no chirp), and also the effect of linear chirp (with uniform apodization) for a given length and height of grating. (For both SMF and Slab Waveguide)

d) What will happen when grating height is varied keeping grating length constant and vice-versa?

(B) When two waveguides are kept side by side, coupling of light occurs from one waveguide to the other. Coupling can be either co-directional or contra-directional and the amount of coupling from one waveguide to another is defined by the coupling coefficient. For such a two- waveguide system placed side by side, define,

a) Why does coupling occur?
b) On what factors do the coupling coefficient depends?
c) What do you mean by beat length of a co-directional coupler?
d) Obtain effective index of the modes of Waveguide 1 and 2 from OptiGrating. (Use data given below).

e) For a co-directional coupler and contra-directional coupler, find out the following:

1. The required grating period for the energy coupling between modes given below

i. Couple TE_0 mode of the 1st waveguide to TE_0 of 2nd waveguide.
ii. Couple TE_0 mode of 1st waveguide to TE_1 of 2nd waveguide.
iii. Couple TE_1 mode of 1st waveguide to TE_0 of 2nd waveguide.

iv. Couple TE_1 mode of 1st waveguide to TE_1 of 2nd waveguide.

2. Find out Beat length for each of the above case (in case of co- directional coupler.

3. Observe power transfer characteristic at wavelength other than 1550nmwhile the grating is designed for coupling at 1550nm.

Waveguide 1	**Waveguide 2**
Cladding Width = 5 µm, n = 1.444 Core Width = 1 µm, n = 3.473 Substrate Width = 5 µm, n = 1.444	Cladding Width = 5 µm, n = 1.444 Core Width = 0.3 µm, n = 3.473 Substrate Width = 5 µm, n = 1.444
Gap between the waveguides = 0.6 µm	

Theory: An optical grating is a device in optics and physics characterized by a periodic arrangement of closely spaced, parallel slits or grooves. These structures interact with incident light, causing it to diffract and form an interference pattern. This diffraction enables the separation of light into its component wavelengths, making optical gratings essential in applications such as spectroscopy, wavelength analysis, and optical communications. By analyzing the diffraction pattern, various wavelengths can be identified and studied, highlighting the importance of optical gratings in fields like photonics and wavelength-selective technologies.

Grating Period: The distance between adjacent grooves or lines on an optical grating, often represented as *d*, is a key parameter that governs how the grating diffracts light. This spacing plays a critical role in determining the angles at which various wavelengths of light are dispersed, directly influencing the grating's ability to separate different wavelengths.

Optical coupling: Optical coupling refers to the process of transferring light between two optical components, such as from a laser to an optical fiber or between waveguides in a photonic integrated circuit. Achieving efficient optical coupling is essential for reducing light loss and ensuring optimal energy transfer between the components.

Co-directional Coupling: In co-directional coupling, light travels in the same direction within two adjacent waveguides or optical paths. As the waveguides are coupled along a specific length, energy from one waveguide is progressively transferred to the other.

Contra-directional Coupling: In contra-directional coupling, light travels in opposite directions within the coupled waveguides or optical paths. This type of coupling typically relies on a periodic structure, such as a grating, to enable the transfer of light between the waveguides.

Beat Length: It refers to the distance over which light transferred from one waveguide to another completes a full cycle of power exchange. In simpler terms, it is the length of the coupler where light initially in one waveguide is fully transferred to the neighboring waveguide, after which it begins to transfer back.

Mathematical Analysis:

(a) For single Mode Fiber:

Cladding Width = 125 µm, n_2 = 1.444, n_1 = 1.4475, V number = 2.419 (Practical Data)

We Know,

$$V = 2\pi/\lambda(n_1^2\text{-}n_2^2)^{1/2}$$

On Substituting values, we get, a = 5.899 µm.

(b) For Slab Waveguide:

Cladding Width = 5 µm, n_2 = 1.444, n_1 = 3.473, V number = 3.14 (Practical Data)

We Know,

$$V = 2\pi/\lambda(n_1^2\text{-}n_2^2)^{1/2}$$

On Substituting values, we get, **a = 0.245 µm.**

Grating Period Calculation:

(a) For single Mode Fiber:

Bragg Wavelength (λ_B) = 2 x Grating Period (Λ) x n_{eff} n_{eff} = 1.45 (practical data)

$1.55 \times 10^{-6} = 2 \times \Lambda \times 1.45$

Λ = 0.534 µm

(b) For Slab Waveguide:

Bragg Wavelength (λ_B) = 2 x Grating Period (Λ) x n_{eff} n_{eff} = 3.25 (practical data)

$1.55 \times 10^{-6} = 2 \times \Lambda \times 3.25$

Λ = 0.2381 µm

<u>Result And Plots:</u>

- **In practical single mode fiber become multi-mode at a = 9.511 micro meter.**
- **In practical single mode fiber become multi-mode at a = 0.264 micro meter.**

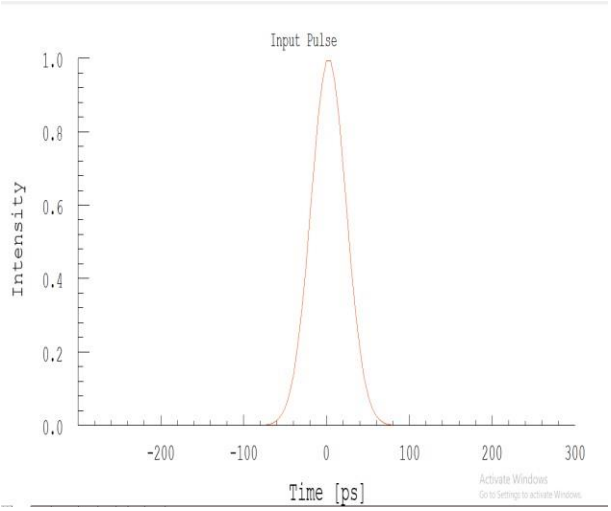

Fig. 1 Mode Profile of SMF.

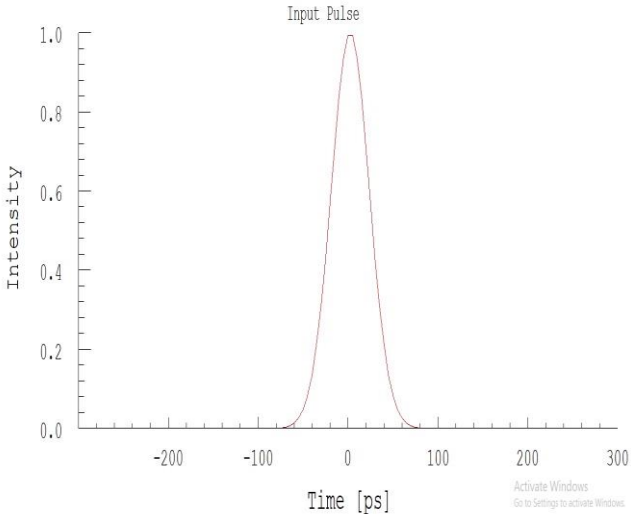

Fig. 2 Mode Profile of Slab Waveguide (SW).

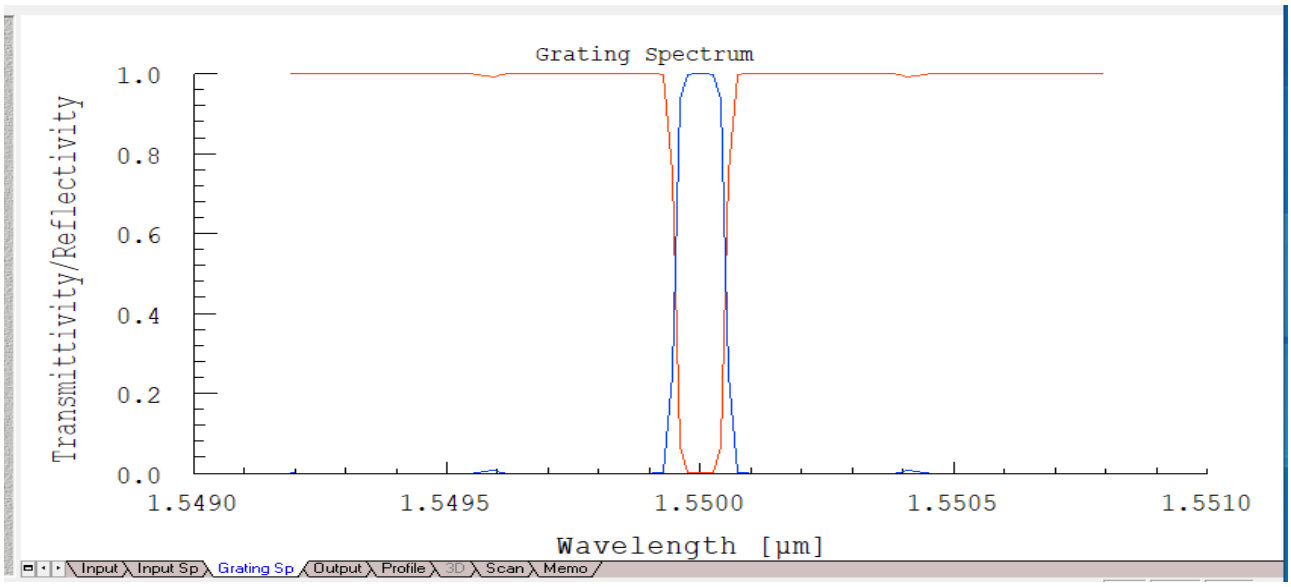

Fig. 3 Gaussian Apodization with No Chirp in SMF.

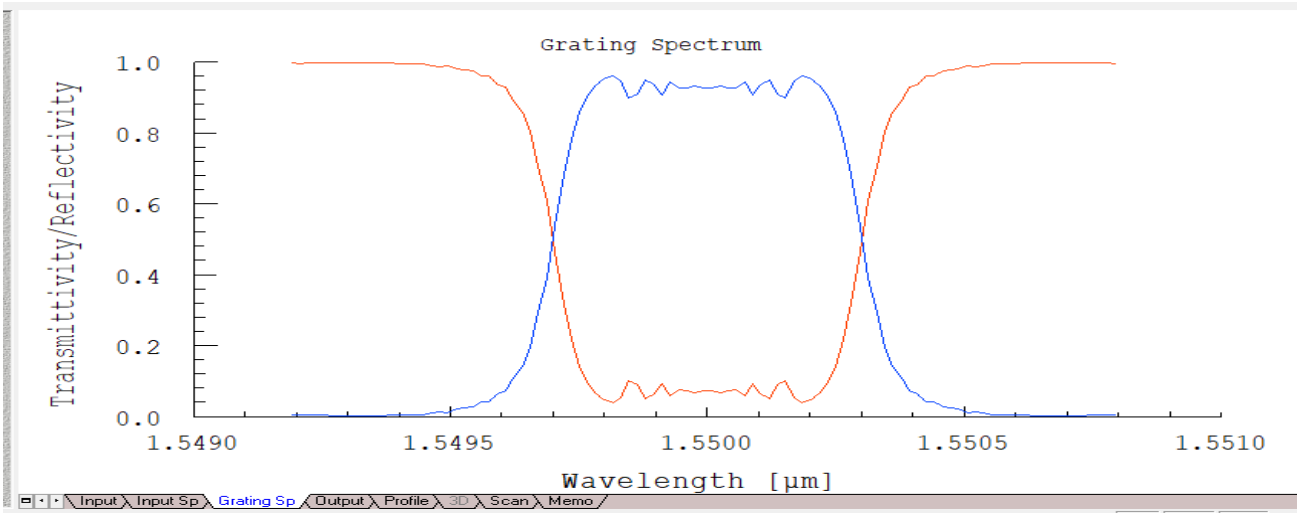

Fig. 4 Uniform Apodization with Linear Chirp in SMF.

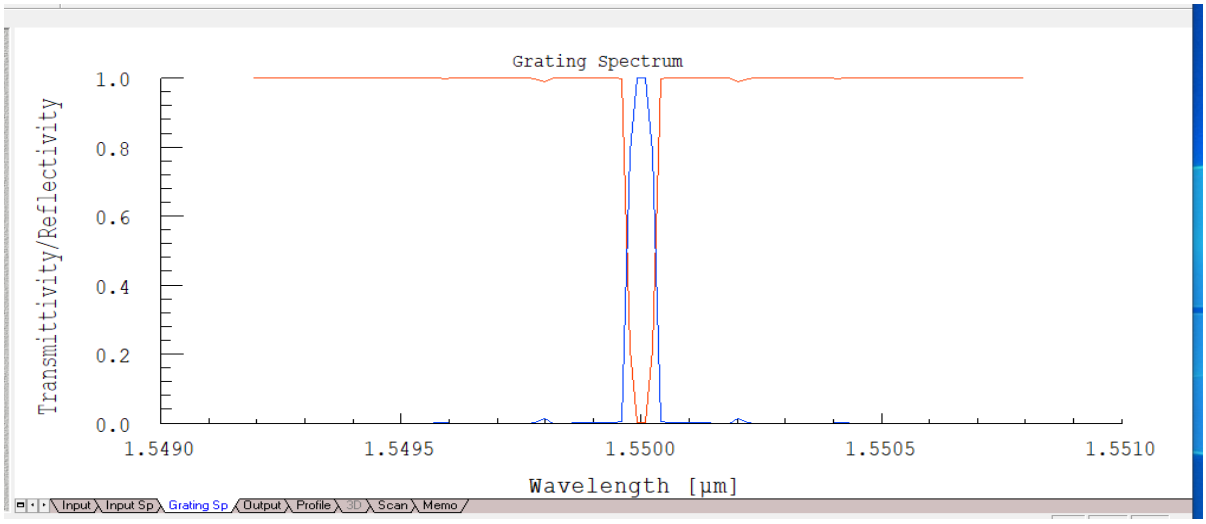

Fig. 5 Gaussian Apodization with No Chirp in SW

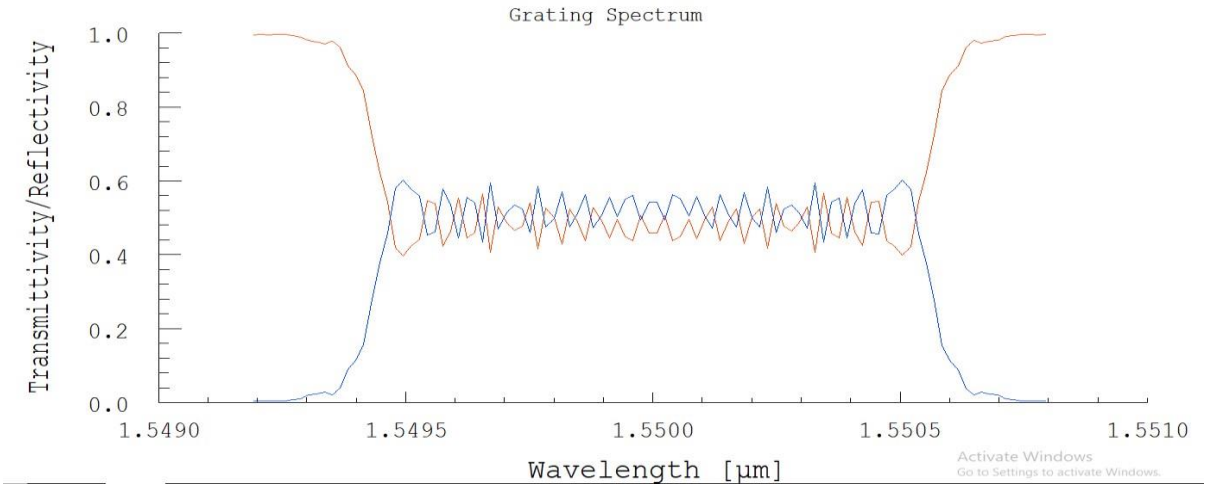

Fig. 6 Uniform Apodization with Linear Chirp in SW.

- **Variation in grating height keeping grating length constant in SMF.**

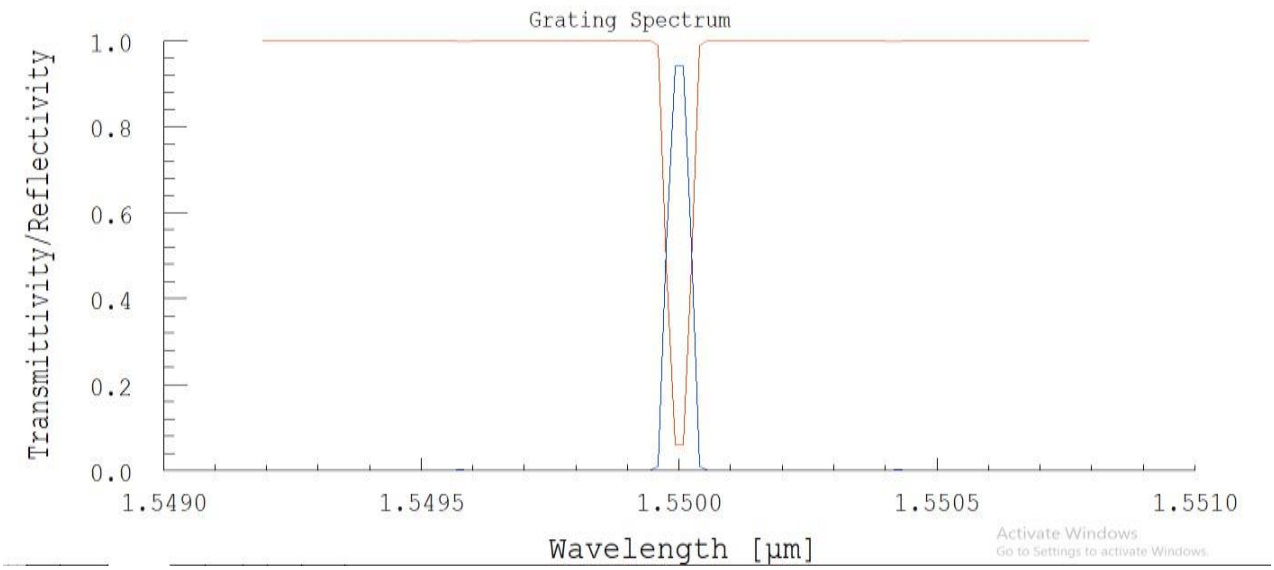

Fig. 7 Grating Length = 50000 µm, Grating height = 0.00005 µm.

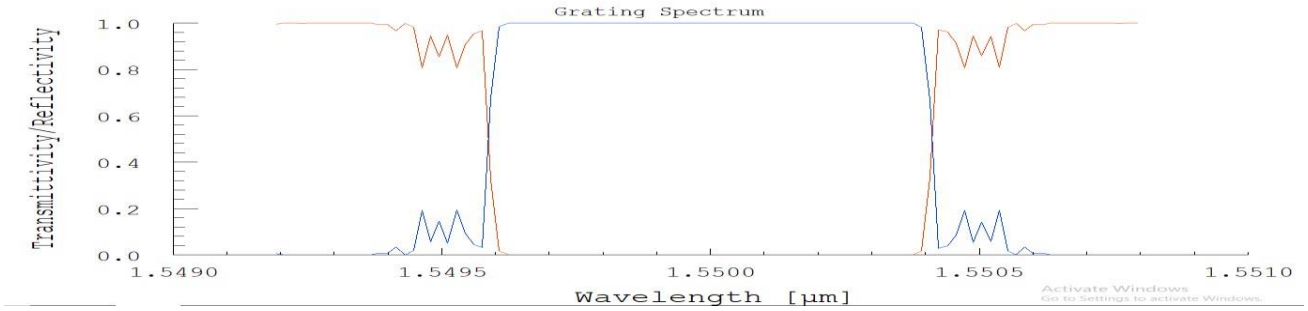

Fig.8 Grating Length = 50000 µm, Grating height = 0.0009 µm.

- **Variation in grating height keeping grating length constant in SW.**

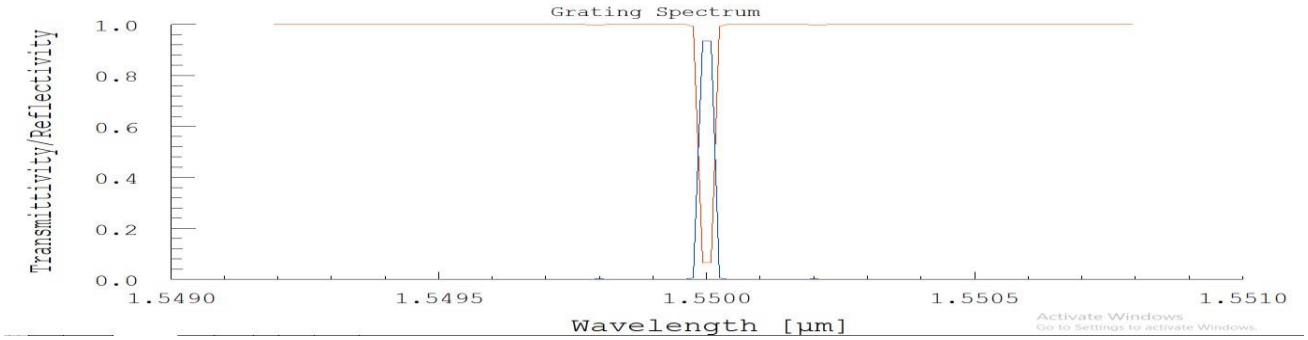

Fig. 9 Grating Length = 50000 µm, Grating height = 0.00005 µm.

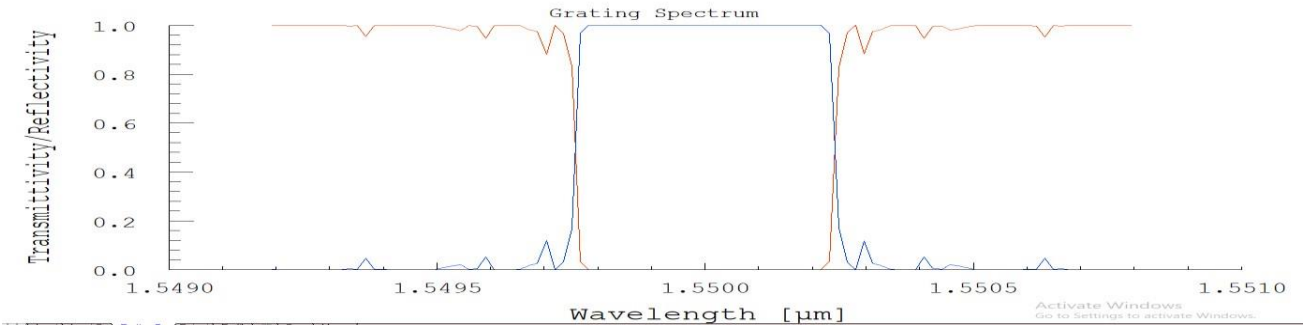

Fig.10 Grating Length = 50000 µm, Grating height = 0.0009 µm.

Fig.11 Effective Indices of the modes of Waveguide 1 and 2.

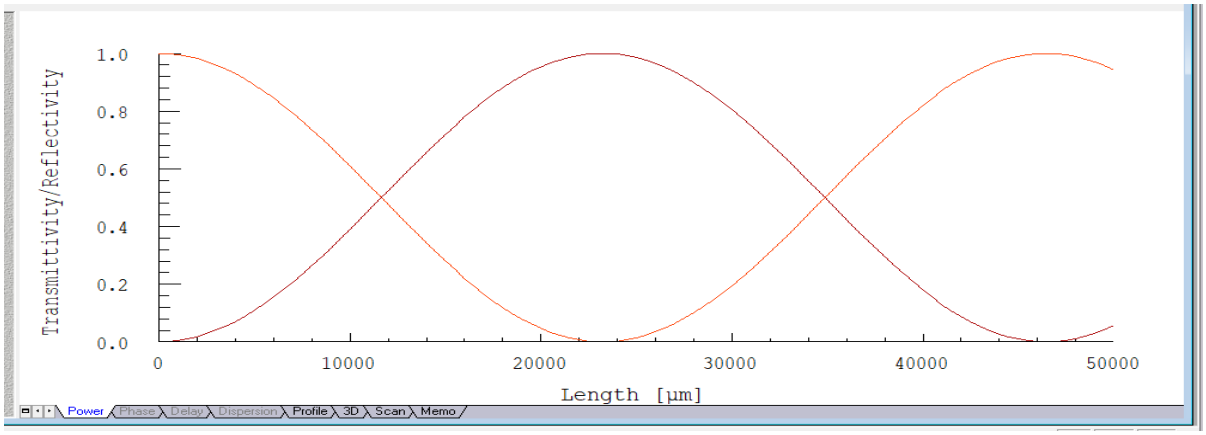

Fig. 12 Co-Directional Coupling from **TE$_0$** mode of the 1st waveguide to **TE$_0$** of 2nd waveguide.

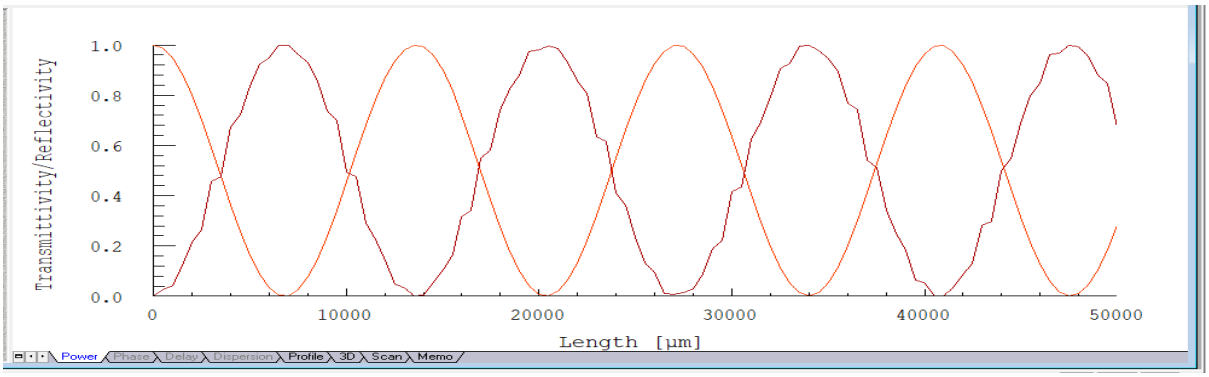

Fig. 13 Co-Directional Coupling from **TE$_0$** mode of the 1st waveguide to **TE$_1$** of 2nd waveguide.

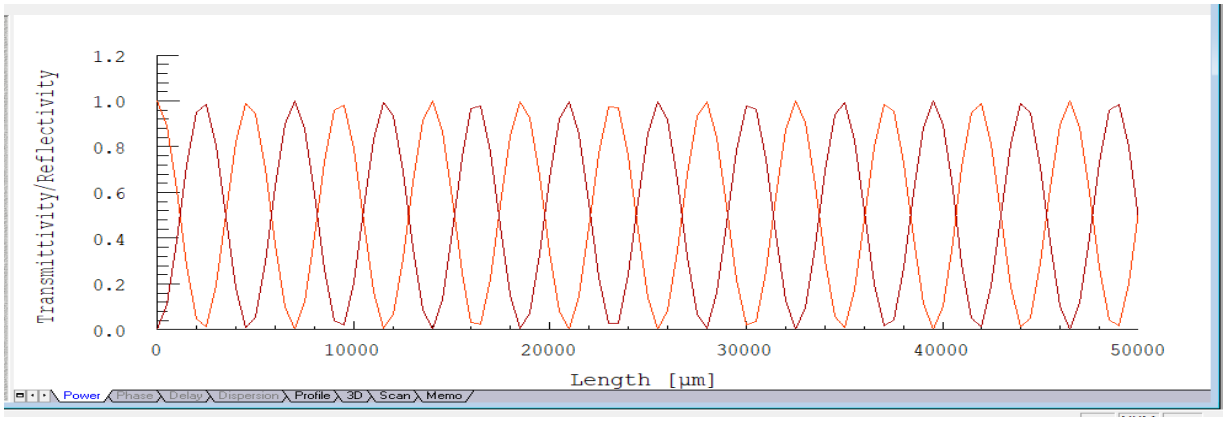

Fig. 14 Co-Directional Coupling from **TE$_1$** mode of the 1st waveguide to **TE$_0$** of 2nd waveguide.

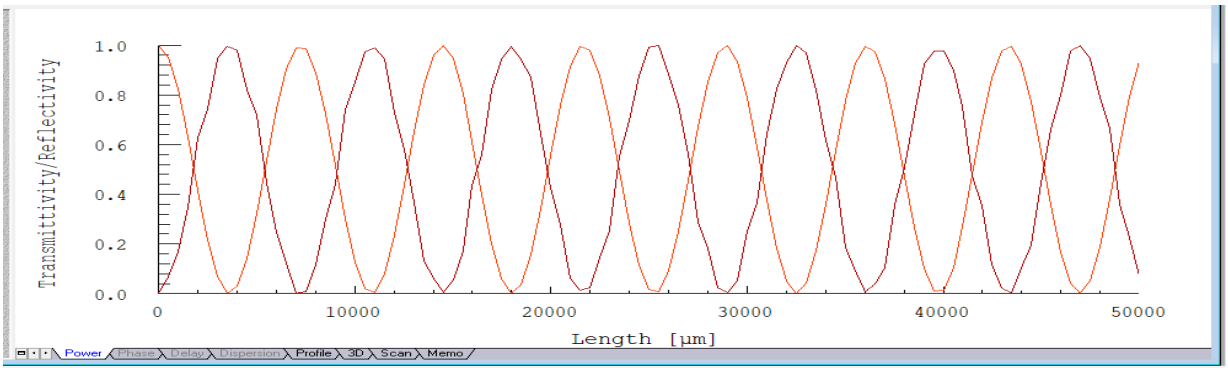

Fig. 15 Co-Directional Coupling from **TE$_1$** mode of the 1st waveguide to **TE$_1$** of 2nd waveguide.

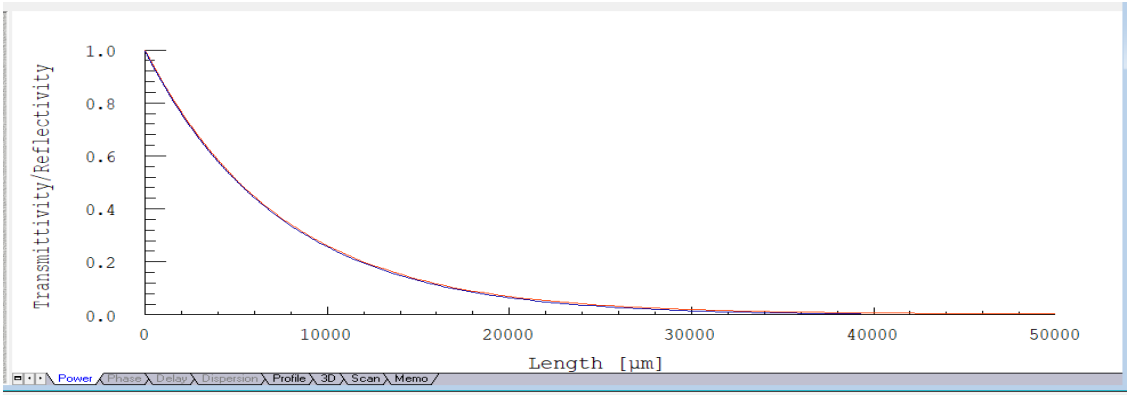

Fig. 16 Contra-Directional Coupling from **TE$_0$** mode of the 1st waveguide to **TE$_0$** of 2nd waveguide.

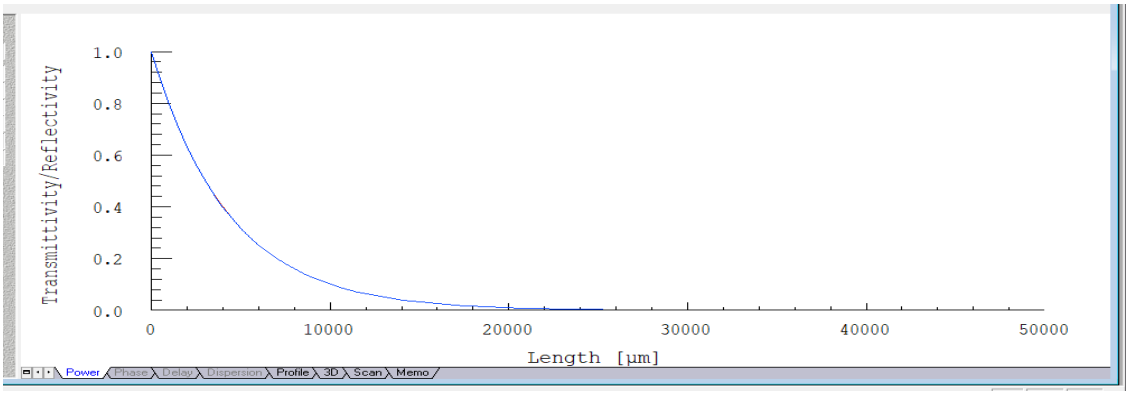

Fig. 17 Contra-Directional Coupling from **TE$_0$** mode of the 1st waveguide to **TE$_1$** of 2nd waveguide.

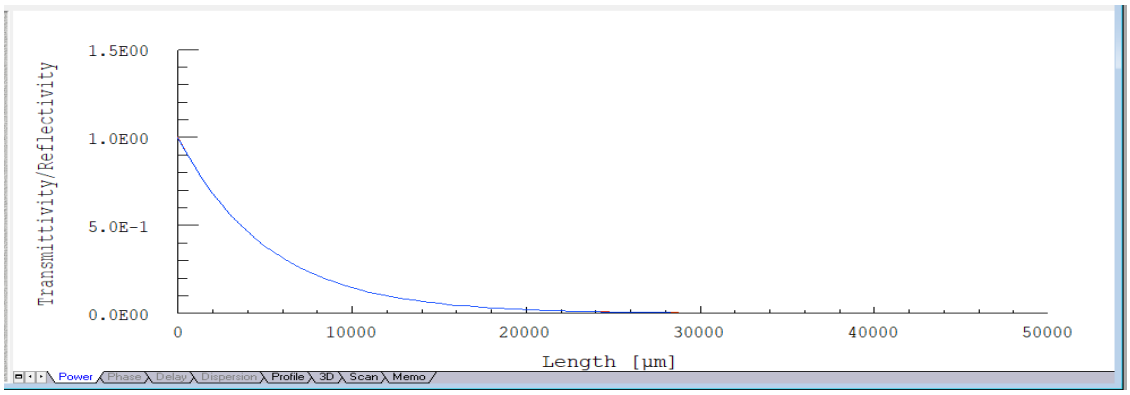

Fig. 18 Contra-Directional Coupling from **TE$_1$** mode of the 1st waveguide to **TE$_0$** of 2nd waveguide.

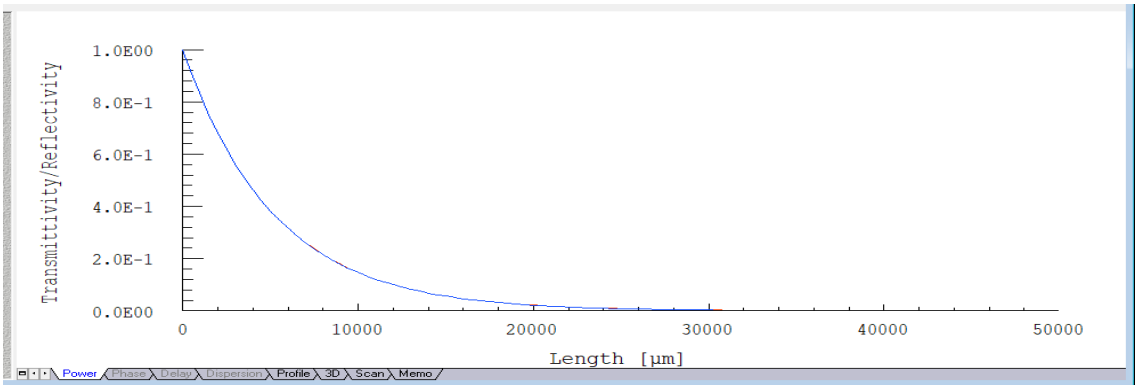

Fig. 19 Contra-Directional Coupling from **TE$_1$** mode of the 1st waveguide to **TE$_1$** of 2nd waveguide.

Observations:

Co-directional coupling from Waveguide 1 (W1) to Waveguide 2 (W2).

Modes	Grating Period (µm)	Beat Length (µm)
TE0-W1 to TE0-W2	4.2849	23500
TE0-W1 to TE1-W2	0.9032	70000
TE1-W1 to TE0-W2	9.6772	2500
TE1-W1 to TE1-W2	1.0238	3500

Contra-directional coupling from Waveguide 1 (W1) to Waveguide 2 (W2).

Modes	Grating Period (µm)
TE0-W1 to TE0-W2	0.2402 at height = 0.5
TE0-W1 to TE1-W2	0.2274 at height = 0.001
TE1-W1 to TE0-W2	0.2417 at height = 0.0002
TE1-W1 to TE1-W2	0.2417 at height = 0.0002

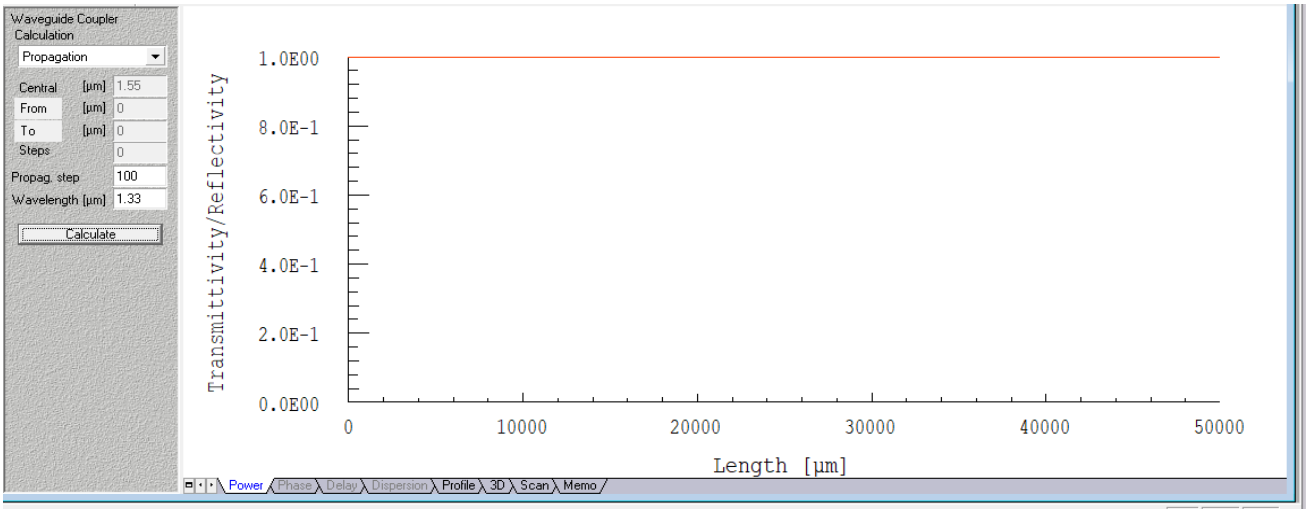

Fig. 20 Power transfer characteristics at wavelength = **1300 nm.**

(B) (a) Why does coupling occur?

Waveguide coupling takes place when two waveguides are positioned close enough for their evanescent fields to overlap, enabling energy transfer between them. This interaction is maximized when the propagation constants of the waveguides are phase-matched, meaning their signal wavelengths or modes are compatible. The strength of the coupling is influenced by the distance between the waveguides and the symmetry of their modes, with shorter distances resulting in stronger interactions. Over a defined coupling length, energy oscillates between the waveguides, allowing efficient power transfer.

b) On what factors do the coupling coefficient depend?

The coupling coefficient depends on several key factors:

1. **Separation Distance**: Closer waveguides enhance evanescent field overlap, strengthening coupling.
2. **Refractive Index Contrast**: A higher contrast between core and cladding refractive indices increases the evanescent field strength.
3. **Waveguide Geometry**: The shape and size influence mode confinement and field overlap.
4. **Wavelength**: Different operating wavelengths affect mode distribution and interaction.
5. **Mode Matching**: Close matching of propagation constants improves coupling efficiency.
6. **Coupling Length**: Longer interaction lengths typically increase coupling.

Conclusion: This experiment successfully demonstrated the design and characterization of a Fiber Bragg Grating (FBG) operating at 1550 nm. The multimode behavior in both fibers and waveguides was investigated, with the core widths determining the transition from single-mode to multimode operation. The effects of Gaussian apodization and chirp on the reflection spectrum were observed, with apodization reducing sidelobes and chirp broadening the bandwidth. Finally, the coupling between adjacent waveguides was analyzed, confirming that the coupling coefficient and beat length depend on mode profiles and waveguide separation.